Subject: Mathematics | **WEEKLY CHAPTER TEST – 04 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Sec A:** 5 MCQs/AR (1M each); **Sec B:** 1 VSAQ (2M) & 3 SA (3M each); **Sec C:** 1 Case Study (4M).

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. If one of the zeroes of a quadratic polynomial of the form $x^2 + ax + b$ is the negative of the other, then it: [1]

(A) no linear term, constant term is negative (B) no linear term, constant term is positive
(C) can have linear term, constant term negative (D) can have linear term, constant term positive

Q2. If the zeroes of the quadratic polynomial $ax^2 + bx + c$, $c \neq 0$ are equal, then: [1]

(A) c and a have opposite signs (B) c and b have opposite signs
(C) c and a have the same sign (D) c and b have the same sign

Q3. If the sum of the zeroes of the quadratic polynomial $f(t) = kt^2 + 2t + 3k$ is equal to their product, then $k =$: [1]

(A) $\frac{2}{3}$ (B) $-\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $-\frac{3}{2}$

Q4. If the product of the zeroes of the quadratic polynomial $f(x) = x^2 - 4x + k$ is 3, then the value of k is: [1]

(A) 3 (B) -3 (C) 4 (D) -4

Q5. **Assertion (A):** A quadratic polynomial having $2 + \sqrt{3}$ and $2 - \sqrt{3}$ as its zeroes is $x^2 - 4x + 1$.
Reason (R): A quadratic polynomial with zeroes α and β is given by $k[x^2 - (\alpha + \beta)x + \alpha\beta]$, where $k \neq 0$. [1]

(A) Both (A) and (R) are true, (R) is correct explanation of (A).
(B) Both (A) and (R) are true, (R) is not correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type Questions (1 × 2 + 3 × 3 = 11 Marks)

Q6. Find the zeroes of $f(x) = 4\sqrt{3}x^2 + 5x - 2\sqrt{3}$ and verify the relationship between the zeroes and coefficients. [2]

Q7. If α and β are the zeroes of the polynomial $f(x) = 2x^2 + 5x + k$ such that $\alpha^2 + \beta^2 + \alpha\beta = \frac{21}{4}$, find the value of k. [3]

Q8. If the sum of squares of zeroes of the quadratic polynomial $f(x) = x^2 - 8x + k$ is equal to 40, find the value of k. [3]

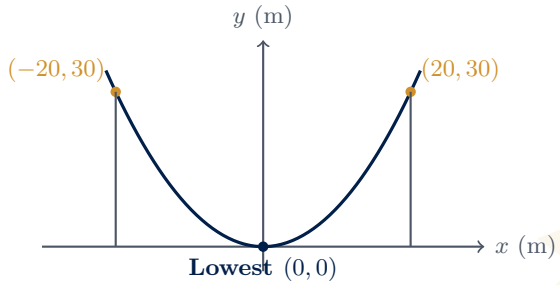
Q9. If α, β are the zeroes of $f(x) = x^2 - p(x + 1) - c$, prove that: [3]

$$(\alpha + 1)(\beta + 1) = 1 - c.$$

SECTION C: Applied Case Study Analysis (1 × 4 = 4 Marks)

Question 10: Case Study Analysis — Suspension Bridge Cable Arch

A suspension bridge has twin towers rising 30 m above the road deck. The main suspension cable hanging between the two towers forms a symmetrical parabolic curve. The cable is anchored to the towers at points $(-20, 30)$ and $(20, 30)$, and its lowest point touches the road deck at the origin $(0, 0)$. The cable's profile is modelled by $y = ax^2 + bx + c$.



Based on the above architectural scenario, answer the following questions:

(i) Using the coordinates of the lowest point and symmetry about the y-axis, determine the values of b and c. **[1]**

(ii) Find the value of coefficient a and write the quadratic equation of the suspension cable curve. **[1]**

(iii) **(a)** Find the height of the suspension cable above the road deck at a horizontal distance of 10 m from the centre. **[2]**

— **OR** —

(b) If vertical suspender rods are installed every 5 m along the deck, determine the length of the suspender rod located at $x = 15$ m. **[2]**

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